

MSE 311 Thermodynamics of materials-Homework 12

Name:

Student ID:

1. (a) Prove the following equation based on the equation of spinodal curve, where $\Omega = \Omega^0 + \Omega^1 T$ (b) Then discuss its first and second differential with respect to X_A . (Note: you must give detailed steps, otherwise you cannot get scores).

$$2(\Omega^0 + \Omega^1 T) X_A (1 - X_A) = RT$$

2. The free energy of mixing of a regular solution is given by ($\Omega = 24,943$ J/mole)

$$\Delta G_{mix} = \Omega X_A X_B + RT (X_A \ln X_A + X_B \ln X_B)$$

- Plot ΔG_{mix} versus X_B at 1400, 1500, and 1600 K.
- Plot $\partial \Delta G_{mix} / \partial X_B$ versus X_B at the same temperatures as (a).
- Determine the critical temperature for this alloy. Show your work.

3. Please draw the Gibbs free energy of mixing curves at various temperatures, and the phase diagram for a binary system which forms regular solid solutions in which $\Omega_s =$

10,000 J and regular liquid solutions in which $\Omega_l = -5,000$ J. The melting temperatures of A and B are, respectively, 800 and 1200 K.

$$\Delta G_{m,(A)}^{\circ} = 8000 - 10T$$

$$\Delta G_{m,(B)}^{\circ} = 12000 - 10T$$

Ps: You should at least draw 5 different temperatures mixing curves and the final phase diagram, at least 6 pictures. You could use the software to simplify your calculate.